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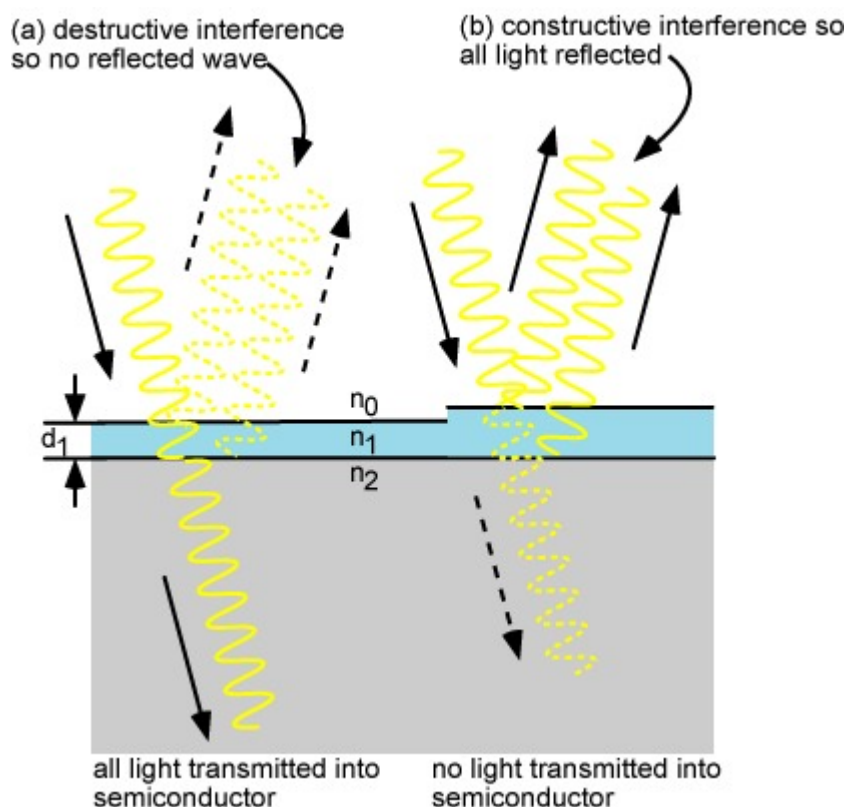
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Anti-Reflection Coatings

Bare silicon has a high surface reflection of over 30%. The reflection is reduced by **texturing** and by applying anti-reflection coatings (ARC) to the surface¹. Anti-reflection coatings on solar cells are similar to those used on other optical equipment such as camera lenses. They consist of a thin layer of dielectric material, with a specially chosen thickness so that interference effects in the coating cause the wave reflected from the anti-reflection coating top surface to be out of phase with the wave reflected from the semiconductor surfaces. These out-of-phase reflected waves destructively interfere with one another, resulting in zero net reflected energy. In addition to anti-reflection coatings, interference effects are also commonly encountered when a thin layer of oil on water produces rainbow-like bands of color.



Use of a quarter wavelength anti-reflection coating to counter surface reflection.

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References

The thickness of the anti-reflection coating is chosen so that the wavelength in the dielectric material is one quarter the wavelength of the incoming wave. For a quarter wavelength anti-reflection coating of a transparent material with a refractive index n_1 and light incident on the coating with a free-space wavelength λ_0 , the thickness d_1 which causes minimum reflection is calculated by:

$$d_1 = \frac{\lambda_0}{4n_1}$$

ARC Thickness Calculator

Wavelength, λ (μm) =

0.6

Refractive index of anti-reflection coating, n_1 :

2

Optimal ARC thickness, d_1 : 0.75 μm

Reflection is further minimized if the refractive index of the anti-reflection coating is the geometric mean of that of the materials on either side; that is, glass or air and the semiconductor. This is expressed by:

$$n_1 = \sqrt{n_0 n_2}$$

ARC Refractive Index Calculator

Refractive index of surrounding material, n_0 :

1.5

Refractive index of semiconductor n_2

3.5

Optimal refractive index of ARC, n_1 : 2.2913

For the reflectance at normal incidence we define a series of parameters: r_1 , r_2 , and θ . The surrounding region has a refractive index of n_0 , the ARC has a refractive index of n_1 and a thickness of t_1 , and the silicon has a refractive index of n_2 .

$$r_1 = \frac{n_0 - n_1}{n_0 + n_1}$$

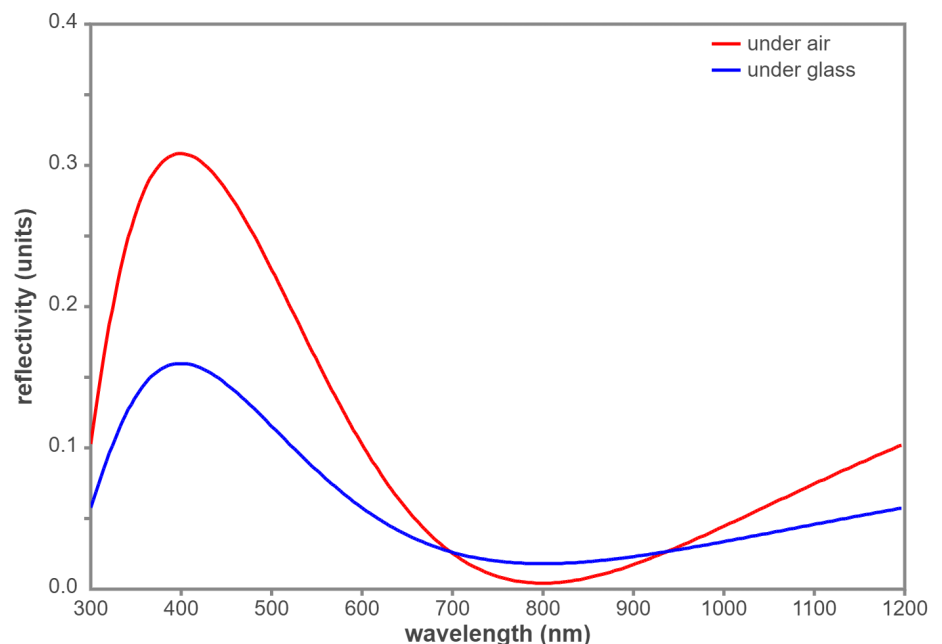
$$r_2 = \frac{n_1 - n_2}{n_1 + n_2}$$

$$\theta = \frac{2\pi n_1 t_1}{\lambda}$$

For a single layer ARC on a substrate the reflectivity is:

$$R = |r^2| = \frac{r_1^2 + r_2^2 + 2r_1 r_2 \cos 2\theta}{1 + r_1^2 r_2^2 + 2r_1 r_2 \cos 2\theta}$$

Single Layer Anti-Reflection Coating (ARC)



Refractive index, n : 2 units

1 3

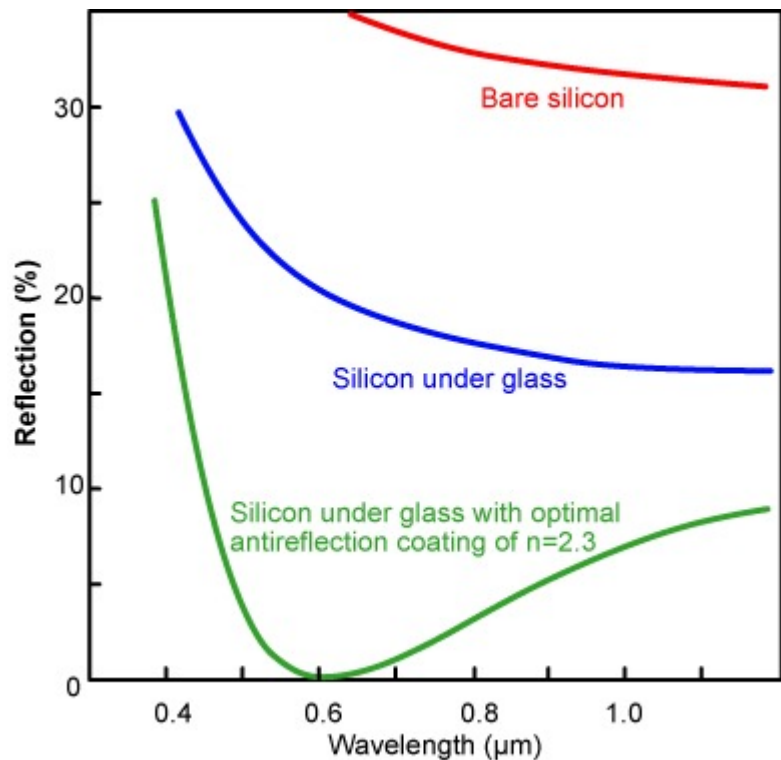
Thickness, t : 100 nm

10 200

The graph shows the effect of a single layer anti-

reflection coating on silicon. Use the sliders to adjust the refractive index and thickness of the layer. For simplicity this simulation assumes a constant refractive index for silicon at 3.5. In reality the **refractive index of silicon** and the coating is a function of wavelength.

While the reflection for a given thickness, index of refraction, and wavelength can be reduced to zero using the equations above, the index of refraction is dependent on wavelength and so zero reflection occurs only at a single wavelength. For photovoltaic applications, the refractive index, and thickness are chosen in order to minimize reflection for a wavelength of $0.6\ \mu\text{m}$. This wavelength is chosen since it is close to the peak power of the solar spectrum.



Comparison of surface reflection from a silicon solar cell, with and without a typical anti-reflection coating.

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1. G. Bauer , “Absolutwerte der optischen Absorptionskonstanten von Alkalihalogenidkristallen im Gebiet ihrer ultravioletten Eigenfrequenzen” , *Annalen der Physik*, vol. 411, no. 4, pp. 434 - 464, 1934.

